

MARTIAL ARTS MOBILE APP

Project Proposal

Android Application | End-to-End Feature Plan

PREPARED FOR

Client | Martial Arts Academy

PREPARED BY

Development Team | 2026

Executive Summary

This document outlines the complete feature set, application flow, recommended technology stack, and development timeline for a **Martial Arts Mobile Application** built for Android. The app is designed to digitise and streamline operations for a martial arts academy — covering student management, batch scheduling, payments, curriculum delivery, and community engagement.

14 Modules	3 User Roles	11 Batches	17 Weeks Timeline	Android Platform
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App Flow

The following table describes the end-to-end user journey through the app, from first launch to daily usage across all three roles — Student, Instructor, and Admin.

Step	Screen / Module	Description
1	App Launch	User opens the Android app
2	Onboarding / Sign In	New student joins and enrolls; existing user logs in
3	Home Dashboard	Central hub — quick access to all modules
4a	Batches & Students	Admin manages 11 batches, assigns instructors, enrolls students
4b	Payments	Student pays fees; admin tracks; receipt auto-generated
4c	Curriculum	Belt/level system with video technique library
4d	Attendance	Instructor marks present/absent; monthly report generated
4e	Feedback	Post-session ratings between instructor and student
5	Community Feed	Events, video posts, notification posts visible to all
6	Progress & Certificate	Belt upgrade triggers auto-certificate; student downloads PDF
7	Admin / Reporting	Owner views batch reports, payments, student analytics
8	Vendor / Shop	Equipment marketplace via Impure vendor integration

Feature List

All features are categorised by module, user role, and development priority. **High** priority items form the MVP (Minimum Viable Product).

Module	Features	User Role	Priority
Batch Management	11 batches, enrol students, batch schedule	Admin	High

Module	Features	User Role	Priority
Student Management	Profile, belt level, progress history	Admin / Student	High
Payments	Fee collection, payment gateway, receipts	Admin / Student	High
Curriculum	Belt/level system, technique library, video content	Instructor / Student	High
Attendance	Mark present/absent, session reports, monthly summary	Instructor	High
Feedback	Post-session ratings, instructor & student reviews	Instructor / Student	Medium
Instructor Management	Assign to batches, profile, schedule	Admin	High
Offline / Online Mode	Switch between offline and online class modes	Admin	Medium
Certificate Generation	Auto-generate on belt upgrade, PDF/image download	Admin / Student	Medium
Events Post	Tournaments, camps, announcements	Admin / Instructor	Medium
Video Post	Tips, highlights, technique demos	Instructor	Medium
Notification Post	Push alerts, reminders, updates	Admin	Medium
Vendor / Shop	Equipment & gear marketplace (Impure vendor)	Admin / Vendor	Low

Recommended Tech Stack

The stack below is selected for reliability, cost-effectiveness, and suitability for the Indian market. Flutter enables a single Android codebase today with an easy path to iOS in the future.

Layer	Technology	Reason
Mobile (Android)	Flutter (Dart)	Single codebase, fast UI, future iOS support
Backend API	Node.js + Express	Lightweight, scalable REST APIs
Database	PostgreSQL	Relational data — students, batches, payments
Authentication	Firebase Auth / JWT	Secure login, OTP support
Cloud Storage	Firebase Storage / S3	Videos, certificates, profile photos
Push Notifications	Firebase Cloud Messaging	Alerts, reminders, event posts
Payment Gateway	Razorpay / Stripe	India-ready, UPI + card support
Hosting	AWS / DigitalOcean	Reliable, scalable cloud infrastructure

Development Timeline

The project is broken into 6 phases over approximately 17 weeks. Each phase delivers working, testable software.

Phase	Duration	Deliverables	Status
Phase 1 Requirements & Design	2 weeks	Wireframes, UI/UX design, DB schema	Planning
Phase 2 Core Features	6 weeks	Auth, batches, students, attendance, payments	Development
Phase 3 Content & Community	4 weeks	Curriculum, video posts, events, notifications	Development
Phase 4 Certificates & Vendor	2 weeks	Certificate generation, vendor/shop module	Development
Phase 5 Testing & QA	2 weeks	Bug fixes, performance, user testing	QA
Phase 6 Launch	1 week	Play Store deployment, go-live support	Launch
TOTAL	~17 weeks	Full Android app — production ready	

Next Steps

- Review this proposal and confirm feature priorities with the development team.
- Finalise the project budget and sign the agreement.
- Begin Phase 1: Requirements gathering and UI/UX wireframes.
- Set up project management board (Jira / Trello) and communication channel.
- Weekly progress calls throughout the development lifecycle.

This proposal is confidential and prepared exclusively for the client. All timelines are estimates and subject to final scope agreement.